

(1001CJA621621001)

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

Test Pattern

JEE(Advanced)

REVIEW TEST

28-02-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	C	B	A	C	A,C	C,D	B,C	B,D
SECTION-II	Q.	1	2	3	4	5	6				
	A.	800.00	3.14	1995.00	0.38	4.80	1.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	B	B	A	C	A,B,C	A,C,D	A,C,D	A
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.05 to 3.09	0.00	6.00	1.80	3.00	8.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	C	C	D	A,B,C	B,C	A,B,D	B,D
SECTION-II	Q.	1	2	3	4	5	6				
	A.	48.00	8.00	1.00	0.14	0.11	3.00				

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JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,C	A,B,C	A,B,D	A,B,C,D	B,C,D	A,C,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	42.00	2.00	1.25	3.80 to 3.90	3.12 to 3.13	2.31				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	4	3	4	6	7	4				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,B	A,B,C,D	A,C,D	A,B,C,D	A,C	A,B,C,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.65 or 5.66	2.66 or 2.67	7.00	1.66 or 1.67	1.00	4.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	7	3	4	6	4	5				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,C,D	A,C,D	B,C	C	A,B,C,D	B				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	2.66 or 2.67	-1.00	1.00	0.50	2.66 or 2.67	1560.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	2	8	4	6	1	4				



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PAPER-1

PART-2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans.(C)
2. Ans.(D)
3. Ans.(B)
4. Ans.(B)
5. Ans.(A)
6. Ans.(C)
7. Ans.(A,B,C)
8. Ans.(A,C,D)
9. Ans.(A,C,D)
10. Ans.(A)
11. Ans.(B,D)
12. Ans.(A,B,D)

SECTION-II

1. Ans.(3.05 to 3.09)
2. Ans.(0.00)
3. Ans.(6.00)

4. Ans.(1.80)

$$C_m = C_v - \frac{R}{x-1}$$

$$C_p - 2R = C_v - \frac{R}{x-1}$$

$$C_v - R = C_v - \frac{R}{x-1}$$

$$x - 1 = 1$$

$$x = 2$$

5. Ans.(3.00)
6. Ans.(8.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

$$\left(\frac{x+1+\frac{1}{x-1}}{1-\frac{x+1}{x-1}} \right) = \tan \left(2 \tan^{-1} \frac{4}{3} \right) = \tan \left(\pi - \tan^{-1} \frac{24}{7} \right)$$

$$\Rightarrow x^2 = \frac{48}{7}$$

$$x = \pm 4\sqrt{\frac{3}{7}}$$

2. **Ans. (C)**

$$\text{Given } f(x+5) \geq f(x) + 5 \quad \dots(1)$$

$$f(x+1) \leq f(x) + 1 \quad \dots(2)$$

from (1) and (2), we can write

$$\begin{aligned} f(x) + 5 &\leq f(x+5) \leq f(x+4) + 1 \\ &\leq f(x+3) + 2 \\ &\leq f(x+2) + 3 \\ &\leq f(x+1) + 4 \\ &\leq f(x) + 5 \end{aligned}$$

$\therefore$ equality exist everywhere

$$\therefore f(x+1) = f(x) + 1 \quad \therefore f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3, \dots, f(2021) = 2021.$$

$$\text{Now } g(x) = f(x) + 5 - x \Rightarrow g(2021) = 5$$

3. **Ans. (D)**

$$([x]+[y]+[z])([x]-[y])^2 + ([y]-[z])^2 + ([z]-[x])^2 = 8$$

possible only when $[x] + [y] + [z] = 4$

$$[x] = 2, [y] = 1, [z] = 1$$

$$\text{or } [x] = 1, [y] = 2, [z] = 1$$

$$[x] = 1, [y] = 1, [z] = 2$$

4. **Ans. (C)**

$$\therefore (1+\sqrt{2})^n = A_n + B_n \sqrt{2}$$

$$\Rightarrow (1-\sqrt{2})^n = A_n - B_n \sqrt{2}$$

$$\text{on solving, } A_n = \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2},$$

$$B_n = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \lim_{n \rightarrow \infty} \sqrt{2} \cdot \frac{1 + \left(\frac{1-\sqrt{2}}{1+\sqrt{2}} \right)^n}{1 - \left(\frac{1-\sqrt{2}}{1+\sqrt{2}} \right)^n} = \sqrt{2}$$

$$\therefore \left(\frac{1-\sqrt{2}}{1+\sqrt{2}} \right)^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\& (A) \rightarrow \text{limit} = 1 \quad (C) \rightarrow \text{limit} = \sqrt{2}$$

$$(B) \rightarrow \text{limit} = 0$$

5. **Ans. (C)**

$$\text{Sol. } \frac{f'(x)}{f(x)} = \frac{{}^nC_0}{{}^nC_0x+1} + \frac{{}^nC_1}{{}^nC_1x+2} + \dots + \frac{{}^nC_n}{{}^nC_nx+n+1}$$

$$\Rightarrow \frac{f'(0)}{f(0)} = \frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \dots + \frac{{}^nC_n}{n+1}$$

$$f'(0) = (n+1)! \left(\frac{2^{n+1}-1}{n+1} \right) = n!(2^{n+1}-1)$$

$$\therefore n!(2^{n+1}-1) = \frac{3}{2} 7!$$

6. **Ans. (D)**

$$p'(x) = e^{(\ell nx)^2} \cdot \left(\frac{2\ell nx}{x} \right)$$

$$\therefore p'(e^{-1}) = -2e^2$$

$$q'(x) = e^{x\ell n(\ell nx)} \cdot \left[\frac{1}{\ell nx} + \ell n(\ell nx) \right]$$

$$q'(e) = 1$$

$$r'(x) = \frac{1-\ell nx}{x^2}, r'(e^2) = -\frac{1}{e^4}$$

$$\therefore \frac{p'(e^{-1}) \cdot r'(e^2)}{q'(e)} = \frac{2}{e^2}$$

7. **Ans. (A,B,C)**

8. Ans. (B,C)

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{\substack{x \rightarrow 1^+ \\ n \rightarrow \infty}} \frac{2 + \frac{x}{x^{2n}} + \frac{2}{x^{2n}}}{1 - \frac{x^2}{x^{2n}} + \frac{3}{x^{2n}}} = 2$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{\substack{x \rightarrow 1^- \\ n \rightarrow \infty}} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 2}$$

$$= \lim_{x \rightarrow 1^-} \frac{x + 2}{-x^2 + 2} = \frac{3}{1} = 3$$

$\therefore \text{LHL} \neq \text{RHL}$

$\therefore$ Discontinuous at $x = 1$.

Similarly

$$\lim_{x \rightarrow 2^+} f(x) = 2; \lim_{x \rightarrow 2^-} f(x) = 2$$

9. Ans. (A,B,D)

$$f(x) = \begin{cases} \pi \log_{\pi}(-x) & x < -\pi \\ -x & -\pi \leq x < 0 \\ x & 0 \leq x \leq \pi \\ \pi \log_{\pi} x & x > \pi \end{cases}$$

$\therefore f(x)$ is continuous everywhere.

$\Rightarrow$ check differentiability at $x = \pi$

$$\Rightarrow \begin{cases} f'(\pi^+) = \pi \cdot \frac{1}{\pi \ln \pi} \\ f'(\pi^-) = 1 \end{cases} \quad \text{Not differentiable}$$

at $x = 0$

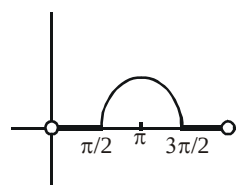
$$\begin{cases} f'(0^+) = 1 \\ f'(0^-) = -1 \end{cases} \quad \text{Not differentiable at } x = 0$$

at $x = -\pi$

$$\begin{cases} f'(-\pi^+) = -1 \\ f'(-\pi^-) = -\frac{1}{\ln \pi} \end{cases} \quad \text{Not differentiable at } x = -\pi$$

10. Ans. (B,D)

$$\cos x [\cos x] = \begin{cases} 0 & 0 < x \leq \frac{\pi}{2} \\ -\cos x & \frac{\pi}{2} < x \leq \pi \\ -\cos x & \pi < x < \frac{3\pi}{2} \\ 0 & \frac{3\pi}{2} \leq x < 2\pi \end{cases}$$



11. Ans. (A,B)

$$\text{Clearly } \lim_{x \rightarrow \alpha^+} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3} = 0$$

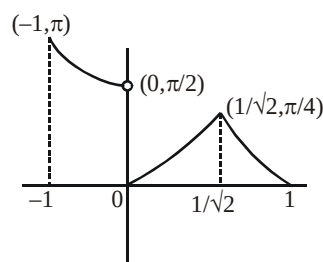
$\therefore$ for existence of limit L.H.L = 0.

$$\Rightarrow \alpha^3 - 1 - (\alpha - 1)^3 = 0$$

$$\Rightarrow \alpha^3 - 1 - (\alpha^3 - 3\alpha^2 + 3\alpha - 1) = 0$$

$$3\alpha^2 - 3\alpha = 0 \Rightarrow \alpha = 0, 1$$

12. Ans. (A,C,D)



$$\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 1$$

SECTION-II

1. Ans. 48.00

2. Ans. 8.00

$f(x)$ is discontinuous at

$x = \pm\sqrt{3}, \pm\sqrt{2}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$ only since $\sin \pi x = 0$ at $x = \pm 1, \pm 2, 0, 3$ are points where $f(x)$ is continuous.

3. Ans. 1.00

$$f(x) = |(2^x - 1) \cdot (3^x - 3)x \cdot (x - 1) \cdot (x - 2)|$$

$f(x)$ is not differentiable at $x = 2$

4. Ans. 0.14

$$S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57}$$

$$= \sum_{k=0}^n \tan^{-1} \frac{1}{k^2 + \frac{63}{49}k + \frac{57}{49}}$$

$$= \sum_{k=0}^n \tan^{-1} \frac{1}{1 + \left(k + \frac{8}{7}\right)\left(k + \frac{1}{7}\right)}$$

$$= \sum_{k=0}^n \left(\tan^{-1} \left(k + \frac{8}{7} \right) - \tan^{-1} \left(k + \frac{1}{7} \right) \right)$$

$$\Rightarrow S_n = \tan^{-1} \left(n + \frac{8}{7} \right) - \tan^{-1} \frac{1}{7}$$

$$S_\infty = \cot^{-1} \frac{1}{7}$$

5. Ans. 0.11

$$x = 1, \quad f(1) + 2f(1) = 1$$

$$\Rightarrow f(1) = \frac{1}{3}$$

$$f(x^2) + 2x^2 f'(x^2) + (x^2 + 1)f'(x) + 2xf(x) = 1$$

$$x = 1$$

$$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$$

$$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$$

$$2xf'(x^2) + 4x^3 f''(x^2) + 4xf'(x^2) + 2xf'(x)$$

$$+ (x^2 + 1)f''(x) + 2f(x) + 2f'(x) = 0$$

$$x = 1$$

$$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1)$$

$$+ 2f(1) + 2f'(1) = 0$$

$$6f''(1) + \frac{2}{3} = 0$$

$$f''(1) = -\frac{1}{9}$$

6. Ans. 3.00

$$f(x) = \begin{cases} ax & ; x < 2 \\ ax^2 - bx + 3 & ; x \geq 2 \end{cases}$$

for $f(x)$ to be continuous $\forall x \in \mathbb{R}$ it should be continuous at $x = 2$

$$f(2^-) = \lim_{x \rightarrow 2^-} ax = 2a$$

$$f(2^+) = f(2) = 4a - 2b + 3$$

$$f(2^-) = f(2) = f(2^+) \Rightarrow 2a = 4a - 2b + 3$$

$$2a - 2b + 3 = 0$$

$$\dots(1)$$

$$f'(x) = \begin{cases} a & ; x < 2 \\ 2ax - b & ; x > 2 \end{cases}$$

$f(x)$ will be differentiable $\forall x \in \mathbb{R}$ if it is differentiable at $x = 2 \Rightarrow f'(2^-) = f'(2^+)$

$$\Rightarrow a = 4a - b$$

$$\Rightarrow 3a - b = 0$$

$$\dots(2)$$

$$\text{By (1) \& (2) } a = \frac{3}{4} \text{ and } b = \frac{9}{4}$$

$$\therefore a + b = 3$$



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PAPER-2

PART-2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans.(A,B)
2. Ans.(A,B,C,D)
3. Ans.(A,C,D)
4. Ans.(A,B,C,D)

Sol. $-\frac{1}{2} \frac{d[A]}{dt} = K[A] \Rightarrow -\frac{d[A]}{dt} = 2K[A]$

$$K' = 2K = 2 \times 1.386 \times 10^{-2} \text{ min.}$$

(i) $K' = \frac{1}{t} \ln \frac{a}{a-x} \Rightarrow 2 \times 1.386 \times 10^{-2}$

$$= \frac{1}{t} \ln \frac{100}{25} \quad t = 50 \text{ min}$$

(ii) $2 \times 1.386 \times 10^{-2} = \frac{1}{t} \ln \frac{100}{(0.1)}, t = 250 \text{ min}$

5. Ans.(A,C)
6. Ans.(A,B,C,D)

SECTION-II

1. Ans.(5.65 or 5.66)
2. Ans.(2.66 or 2.67)
3. Ans.(7.00)
4. Ans.(1.66 or 1.67)
5. Ans.(1.00)
6. Ans.(4.00)

SECTION-III

1. Ans.(7)
2. Ans.(3)
3. Ans.(4)

4. Ans. (6)

Sol. Molarity of 1st solution

$$= \frac{0.2 \times M}{0.2M + 18 \times 0.8} \times 1000$$

$$d_1$$

Molarity of 2nd solution = $\frac{0.1}{0.1M + 0.9 \times 18} \times 1000$

$$d_2$$

$$\Rightarrow d_2 = 0.8d_1$$

$$\frac{1}{2} \times \frac{0.2 \times 1000}{0.2M + 18 \times 0.8} \times d_1 = \frac{0.1 \times 0.8d_1}{0.1M + 0.9 \times 18}$$

$$\Rightarrow \frac{5}{0.2M + 18 \times 0.8} = \frac{4}{0.1M + 0.9 \times 18}$$

$$M = 78$$

5. Ans.(4)

6. Ans.(5)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,C,D)**

$$OM = OC + CM$$

$$\Rightarrow x = r\sqrt{2} + r$$

$$\Rightarrow \frac{x}{r} = \sqrt{2} + 1 \Rightarrow \frac{r}{x} = \sqrt{2} - 1$$

$$\Rightarrow \frac{dx}{dr} = \sqrt{2} + 1 \text{ and } \frac{dr}{dx} = \sqrt{2} - 1$$

$$\text{and } \frac{d^2x}{dr^2} = 0.$$

Now

$$\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 = (\sqrt{2} + 1)^2 + (\sqrt{2} - 1)^2 = 2(2 + 1) = 6$$

$$\& \frac{x+r}{x-r} = \frac{\sqrt{2}+2}{\sqrt{2}} \text{ and } \frac{r-x}{r+x} = \frac{\sqrt{2}-2}{\sqrt{2}}$$

$$\Rightarrow \frac{x+r}{x-r} + \frac{r-x}{r+x} = 2$$

2. **Ans. (A,C,D)**

Sol.

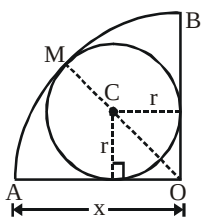
$$f(|x|) = \begin{cases} |x| - 2, & 0 \leq |x| \leq 3 \end{cases}$$

$$\Rightarrow f(|x|) = \begin{cases} -x - 2, & -3 \leq x \leq 0 \\ x - 2, & 0 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(|x|)] = \begin{cases} [-x] - 2, & -3 \leq x \leq 0 \\ [x] - 2, & 0 < x \leq 3 \end{cases}$$

$$\text{Also } |f(x)| = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2 - x, & 0 < x \leq 2 \\ x - 2, & 2 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(x)] = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2 + [-x], & 0 < x \leq 2 \\ [x] - 2, & 2 < x \leq 3 \end{cases}$$



$$\text{So } [f(|x|)] + [f(x)] = \begin{cases} [-x], & -3 \leq x \leq 0 \\ [x] + [-x], & 0 < x \leq 2 \\ 2[x], & 2 < x \leq 3 \end{cases}$$

Now, options can be checked directly without further simplification.

3. **Ans. (B,C)**

Apply continuity at $x = 1$,

$$2a + b = 2$$

Apply continuity at $x = 3$,

$$2a + 3b = 0$$

$$\text{from (1) \& (2) } a = \frac{3}{2} \text{ and } b = -1$$

4. **Ans. (C)**

$$(\theta_1 + \theta_2) \left(\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} \right) = 1$$

$$\cos^4 x + \sin^4 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = \theta$$

$$1 - 2\sin^2 x \cos^2 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = 0$$

$$\left(\sqrt{\frac{\theta_2}{\theta_1}} \cos^2 x - \sqrt{\frac{\theta_1}{\theta_2}} \sin^2 x \right)^2 = 0$$

$$\tan^2 x = \frac{\theta_2}{\theta_1}$$

$$\theta_2 = \theta_1 \tan^2 x$$

$$\frac{d\theta_2}{d\theta_1} = \tan^2 x$$

5. **Ans. (A,B,C,D)**

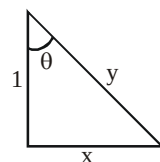
$$y \sin \theta = x$$

$$y \cos \theta = 1$$

$$(y^n - x^n) = \frac{1 - \sin^n \theta}{\cos^n \theta}$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = 0, \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 \theta}{\cos^2 \theta} = 1$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 \theta}{\cos^3 \theta} = \infty \Rightarrow \text{Limit does not exist.}$$



6. Ans. (B)

$$\begin{aligned} \text{Sol. } \lim_{h \rightarrow 0} \frac{(\log(2+h) - \log 2)3 \cdot (4^h - 1 - h)}{((8+h)^{1/3} - (4+3h)^{1/2}) \cdot \sin \pi(1+h)} \\ = \lim_{h \rightarrow 0} \frac{3 \cdot \log\left(1 + \frac{h}{2}\right) \cdot (4^h - 1 - h)}{2 \left[\left(1 + \frac{h}{8}\right)^{1/3} - \left(1 + \frac{3h}{4}\right)^{1/2} \right] (-\sin \pi h)} \\ = \lim_{h \rightarrow 0} \frac{3 \cdot \left[\log\left(1 + \frac{h}{2}\right) \cdot \frac{h}{2} \right] \cdot (4^h - 1 - h)}{2 \left[\left(1 + \frac{h}{24} + \dots\right) - \left(1 + \frac{3h}{8}\right) \right] \cdot \left(\frac{-\sin \pi h}{\pi h} \right) \pi h} \end{aligned}$$

Using standard limit

$$\lim_{h \rightarrow 0} \frac{\left(\frac{3h}{2}\right)(4^h - 1 - h)}{2\left(\frac{h}{3}\right)(\pi h)} = \lim_{h \rightarrow 0} \frac{9}{4\pi} \left(\frac{4^h - 1 - h}{h} \right)$$

Using LH rule

$$\lim_{h \rightarrow 0} \left(\frac{9}{4\pi} \right) \left(\frac{4^h \log 4 - 1}{1} \right)$$

Taking limit $\left(\frac{9}{4\pi} \right) (\log 4 - 1)$

SECTION-II

1. Ans. (2.66 or 2.67)

$$\text{Sol. } \lim_{x \rightarrow -\infty} \sqrt{x^2 - \frac{4x}{3}} + x + 2$$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - \frac{4x}{3} - (x+2)^2}{\sqrt{x^2 - \frac{4x}{3}} - (x+2)} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{-\frac{16}{3}x - 4}{|x| \sqrt{1 - \frac{4}{3x}} - (x+2)}$$

$$\lim_{x \rightarrow -\infty} \frac{x \left(-\frac{16}{3} - \frac{4}{x} \right)}{x \left[-\sqrt{1 - \frac{4}{3x}} - \left(1 + \frac{2}{x} \right) \right]}$$

$$\lim_{x \rightarrow -\infty} -\frac{16}{3(-2)} = \frac{8}{3}$$

2. Ans. -1.00

Using L'Hospital :

$$L = \lim_{x \rightarrow \frac{3\pi}{4}} \frac{4(2 \sin x \cos^2 x - \sin^3 x) + \sin x + \cos x}{\cos x - \sin x} = -1$$

3. Ans. 1.00

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\sin^{-1}(h^2 + h^3) - h}{h} + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2 + h^3) (h^2 + h^3)}{h^2 (h^2 + h^3)} = 1$$

4. Ans. 0.50

Sol.

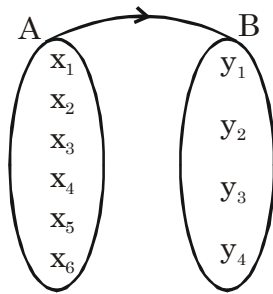
$$T_n = \tan^{-1} \left(\frac{\frac{n+1}{2} - \frac{n}{2}}{1 + \frac{n}{2} \cdot \frac{n+1}{2}} \right) = \tan^{-1} \left(\frac{n+1}{2} \right) - \tan^{-1} \frac{n}{2}$$

5. Ans. (2.66 or 2.67)

$$\text{Sol. } f(x) = \begin{cases} 4x+1; & -\infty < x \leq \frac{1}{3} \\ 2+x; & \frac{1}{3} < x \leq \frac{2}{3} \\ 4-2x; & \frac{2}{3} < x < \infty \end{cases}$$

$$\therefore \text{ Clearly } f_{\max} \left(x = \frac{2}{3} \right) = \frac{8}{3}.$$

6. Ans. 1560.00



Groups 1, 1, 1, 3 and 1, 1, 2, 2

$$\frac{6! \times 4!}{3!(1!)^3 \times 3!} + \frac{6!(4!)}{(1!)^2(2!)^2} \times \frac{1}{(2!)(2!)} = 1560$$

SECTION-III

1. Ans. 2

Sol. $f(x) + f(y) = f(x)f(y) + f(xy)$

partial differentiation wrt x

$$f'(x) = f'(x)f(y) + f'(xy) \cdot (y)$$

put x = 1

$$-2 = -2f(y) + f'(y)y$$

$$\text{get } f(y) = 1 - y^2 \Rightarrow f(x) = 1 - x^2$$

2. Ans. 8

Sol. $\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2 - 3x^3}{3x^4} \right) \cdot 2 = 4$

$$\Rightarrow f(x) = x - 2x^2 + 3x^3 + 6x^4$$

$$\Rightarrow \sum_{i=0}^4 a_i = 1 - 2 + 3 + 6 = 8$$

3. Ans. 4

Required limit is $\left. \frac{d}{dx} (h^{-1}(x)) \right|_{x=1}$

$$\because f^{-1}(x) = x - 1; g^{-1}(x) = x^3 + x + 1$$

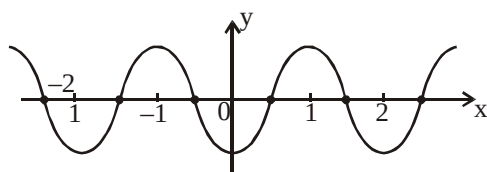
Also $h(x) = g(f(x))$

$$\Rightarrow h^{-1}(x) = f^{-1}(g^{-1}(x)) = x^3 + x$$

$$\Rightarrow \frac{d}{dx} (h^{-1}(x)) = 3x^2 + 1$$

4. Ans. 6

$$\lim_{x \rightarrow \infty} P(x) \rightarrow \infty \text{ and } \lim_{x \rightarrow (-\infty)} P(x) \rightarrow \infty.$$



$P(0) = -1$ and $P(x)$ is even this $P(1) = P(-1) = 1$ & $P(2) = P(-2) = -1$.

Hence, from IVT minimum no. of zeroes are 6.

5. Ans. 1

$$f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} \quad (f(0) = 0)$$

$$f'(x) = 1 + x + x^2 + x^3 + x^4 \quad (f'(0) = 1)$$

$$f''(x) = 1 + 2x + 3x^2 + 4x^3 \quad (f''(0) = 1)$$

Now $f(g(x)) = x$

Diff. w.r.to x

$$f'(g(x))g'(x) = 1 \quad \dots(i)$$

again Diff. w.r.to x

$$f''(g(x))(g'(x))^2 + g''(x)f'(g(x)) = 0$$

$$x = 0$$

$$f''(g(0))(g'(0))^2 + g''(0)f'(g(0)) = 0 \quad \dots(ii)$$

$$\text{as } f(0) = 0 \Rightarrow g(0) = 0$$

$$\text{and from (i) } g'(0) = \frac{1}{f'(0)}$$

now from (ii)

$$f''(0) \left(\frac{1}{f'(0)} \right)^2 + g''(0)f'(0) = 0$$

$$1 \cdot 1^2 + g''(0) \cdot 1 = 0$$

$$g''(0) = -1$$

$$|g''(0)| = 1$$

6. Ans. 4

